Anirban Chowdhury

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Employment

2024 – **Staff Research Scientist,** IBM Research
— research in quantum algorithms and complexity

2020 – 2024 **Postdoctoral Fellow,** Institute for Quantum Computing, University of Waterloo
— research in quantum algorithms and complexity

2019 – 2020 **Postdoctoral Researcher,** Université de Sherbrooke
— research in quantum algorithms for simulating many-body physics

Education

2013 – 2019 **PhD** in Physics, University of New Mexico
Dissertation: *Quantum algorithms with applications to simulating physical systems.*
Advisor: Rolando Somma

2013 – 2016 **MS** in Physics, University of New Mexico
CGPA: 3.67/4.00

2008 – 2013 **BS-MS**, Indian Institute of Science Education and Research Pune
Thesis: Quantum measurements with post-selection
CGPA 8.8/10

Publications

Published Articles

1. Adam Bene Watts, **Anirban Chowdhury**, Aidan Epperly, J William Helton, and Igor Klep. "Relaxations and exact solutions to quantum Max Cut via the algebraic structure of swap operators". In: *Quantum* 8 (May 2024), p. 1352. https://quantum-journal.org/papers/q-2024-05-22-1352/.
2. Sergey Bravyi, **Anirban Chowdhury**, David Gosset, Vojt ěch Havlí ček, and Guanyu Zhu. "Quantum Complexity of the Kronecker Coefficients". In: *PRX Quantum* 5 (1 Feb. 2024), p. 010329. 10.1103/PRXQuantum.5.010329.
3. Sergey Bravyi, **Anirban Chowdhury**, David Gosset, and Pawel Wocjan. "Quantum Hamiltonian complexity in thermal equilibrium". In: *Nature Physics* 18.11 (Oct. 2022), pp. 1367–1370. https://www.nature.com/articles/s41567-022-01742-5.
4. **Anirban N Chowdhury**, Rolando D Somma, and Yiğit Subaşı. "Computing partition functions in the one-clean-qubit model". In: *Physical Review A* 103.3 (Mar. 2021), p. 032422. https://journals.aps.org/pra/abstract/10.1103/PhysRevA.103.032422.
5. **Anirban Narayan Chowdhury** and Rolando Diego Somma. "Quantum algorithms for Gibbs sampling and hitting-time estimation". In: *Quantum Information and Computation* 17.1&2 (Jan. 2017), pp. 0041–0064. https://www.rintonpress.com/xxqic17/qic-17-12/0041-0064.pdf.

6. **Anirban N Chowdhury**, Mandar Patil, Daniele Malafarina, and Pankaj S Joshi. "Circular geodesics and accretion disks in the Janis-Newman-Winicour and gamma metric spacetimes". In: *Physical Review D* 85.10 (May 2012), p. 104031. https://journals.aps.org/prd/abstract/10.1103/PhysRevD.85.104031.

Preprints

1. **Anirban N Chowdhury**, Samuel O Scalet, and Kunal Sharma. *Dissipative microcanonical ensemble preparation from KMS-detailed balance*. Apr. 2026. https://arxiv.org/abs/2604.19973.
2. Arkopal Dutt, **Anirban Chowdhury**, Kristan Temme, and Hari Krovi. *Simulating dynamics of RLC circuits with a quantum differential-algebraic equations solver*. Apr. 2026. https://arxiv.org/abs/2604.26945.
3. David Layden, Ryan Sweke, Vojtěch Havlíček, **Anirban Chowdhury**, and Kirill Neklyudov. *Wavefunction Flows: Efficient Quantum Simulation of Continuous Flow Models*. Oct. 2025. https://arxiv.org/abs/2510.08462.
4. William Kirby, Yizhi Shen, Daan Camps, **Anirban Chowdhury**, Katherine Klymko, and Roel Van Beeumen. *Quantum Krylov Algorithm for Szegö Quadrature*. Sept. 2025. https://arxiv.org/abs/2509.19195.
5. Annie Ray, Esha Swaroop, Ningping Cao, Michael Vasmer, and **Anirban Chowdhury**. *Quasiprobabilistic imaginary-time evolution on quantum computers*. May 2025. https://arxiv.org/abs/2505.06343.
6. Samuel O Scalet, Angela Capel, **Anirban N Chowdhury**, Hamza Fawzi, Omar Fawzi, Isaac H Kim, and Arkin Tikku. *Classical Estimation of the Free Energy and Quantum Gibbs Sampling from the Markov Entropy Decomposition*. Apr. 2025. https://arxiv.org/abs/2504.17405.
7. Mason L Rhodes, Sam Slezak, **Anirban Chowdhury**, and Yiğit Subaşı. *Quantum Approximate Counting with Additive Error: Hardness and Optimality*. Nov. 2024. https://arxiv.org/abs/2411.02602.
8. **Anirban Chowdhury**, Ewout van den Berg, and Pawel Wocjan. *Controlization Schemes Based on Orthogonal Arrays*. July 2024. https://arxiv.org/abs/2407.09382.
9. Chen-Fu Chiang, **Anirban Chowdhury**, and Pawel Wocjan. *Space-efficient Quantization Method for Reversible Markov Chains*. June 2022. https://arxiv.org/abs/2206.06886.
10. **Anirban N Chowdhury**, Guang Hao Low, and Nathan Wiebe. *A Variational Quantum Algorithm for Preparing Quantum Gibbs States*. Jan. 2020. https://arxiv.org/abs/2002.00055.
11. **Anirban Narayan Chowdhury**, Yigit Subasi, and Rolando D. Somma. *Improved implementation of reflection operators*. Mar. 2018. https://arxiv.org/abs/1803.02466.

Patents

- Guang Hao Low, Nathan Wiebe and **Anirban Chowdhury**, *Variational quantum Gibbs state preparation*. Document ID: US 20200349457 A1, Nov 2020

Talks

Invited Seminars

1. *Relaxations to Quantum Max Cut from the Algebraic Structure of Swap Operators*. University of Cambridge, Cambridge, UK Aug. 2024.
2. *Relaxations to Quantum Max Cut from the Algebraic Structure of Swap Operators*. Perimeter Institute, Waterloo, Canada Aug. 2024.
3. *Quantum Counting and the Symmetric Group*. London TQFT Journal Club. Queen Mary University of London (remote), London, UK Apr. 2024.
4. *Quantum Counting and the Symmetric Group*. Institute of Pure and Applied Mathematics (IPAM), UCLA (remote), Los Angeles, USA Nov. 2023.
5. *Approximating Dense Quantum Hamiltonians Using Convex Relaxations*. URA Seminar. University of Waterloo, Waterloo, Canada June 2023.
6. *Relaxations to Local Hamiltonian Problems*. BIRS Workshop on Fundamental Limitations to Quantum Computing. Banff, Canada June 2023.
7. *Quantum Hamiltonian Complexity in Thermal Equilibrium*. Indian Institute of Science Education and Research Pune, Pune, India May 2023.
8. *Quantum Simulation and Complexity of Thermal Physics*. Virginia Tech, Blacksburg, USA Feb. 2023.
9. *Quantum Computing for Quantum Physics and Beyond*. École de Technologie Supérieure (ETS), Montreal, Canada Dec. 2022.
10. *Simulating Quantum Many-Body Physics in Thermal Equilibrium*. Technology Innovation Institute. Abu Dhabi, UAE Mar. 2022.
11. *Simulation of Dense Quantum Hamiltonians Using Convex Relaxations*. Los Alamos National Laboratory (remote), Los Alamos, USA Feb. 2022.
12. *Simulation of Dense Quantum Hamiltonians Using Convex Relaxations*. University of New Mexico (remote), Albuquerque, USA Feb. 2022.
13. *Computing Partition Functions with One Clean Qubit*. University of Bristol (remote), Bristol, UK Mar. 2021.
14. *Simulating Thermal Physics on Quantum Computers*. University of Waterloo, Waterloo, Canada Mar. 2021.
15. *Quantum Algorithms Using a Linear Combination of Unitaries*. Los Alamos National Laboratory, Los Alamos, USA Aug. 2019.

Conference Talks

1. *Quantum and Classical Algorithms for Trace Estimation*. Southwest Quantum Information and Technology Workshop. Tucson, USA Oct. 2022.
2. *On the Complexity of Quantum Partition Functions*. Theory of Quantum Computation, Communication and Cryptography (TQC). Urbana–Champaign, USA July 2022.
3. *On the Complexity of Quantum Partition Functions*. Quantum Information Processing (QIP). Pasadena, USA Mar. 2022.
4. *Simulating Quantum Many-Body Physics in Thermal Equilibrium*. Institute for Fundamental Study. Phitsanulok, Thailand Mar. 2022.
5. *Classical and Quantum Algorithms for Estimating Traces and Partition Functions*. IPAM Workshop on Quantum Numerical Linear Algebra. Los Angeles, USA Jan. 2022.

6. *Computing Partition Functions with One Clean Qubit*. Theory of Quantum Computation, Communication and Cryptography (TQC). Riga, Latvia June 2020.

7. *Simulating Thermal Physics on Quantum Computers*. INTRIQ Fall Meeting. Bromont, Canada Nov. 2019.

8. *Improved Implementation of Reflection Operators*. American Physical Society March Meeting. Boston, USA Mar. 2019.

9. *Quantum Algorithms Using Linear Combination of Unitaries*. Southwest Quantum Information and Technology Workshop. Tucson, USA Feb. 2018.

10. *Quantum Algorithms for Hitting-Time Estimation*. American Physical Society March Meeting. Baltimore, USA Mar. 2016.

Professional Service

Peer Review

- Journals: Nature Physics, npj Quantum Information, Nature Communications, Quantum, Physical Review X, PRX Quantum, Physical Review A, ACM Transactions in Quantum Computing, New Journal of Physics, Journal of the ACM
- Conferences: Quantum Information Processing (QIP) 2022, 2023 and 2025, Theory of Quantum Computation, Communication and Cryptography (TQC) 2020, 2022, 2023, 2025, Foundations of Computer Science (FOCS) 2021, International Colloquium on Automata, Languages and Programming (ICALP 2023), Symposium on Theory of Computer Science (STOCS) 2025

Program Committee

- PC member: QIP 2025, Quantum Computing Theory in Practice (QCTiP) 2025

Other

- Co-organizer: Perimeter Institute Quantum Discussions seminar series 2022

Skills

- Coding: Python, LaTeX, Html, css,
- Languages: English, Hindi, Bengali.

Teaching and Mentorship

- 2024: Mentored Samuel Scalet as an intern at IBM on aspects of dissipative algorithms for quantum state preparation.
- 2022, 2023: Co-supervised two Masters research essays as part of the Perimeter Scholars International program.
- 2017: Graded and assisted students with homework assignments for graduate physics courses.
- 2013-2014: Instructed and supervised students in undergraduate physics lab courses.

Past Employment

2018 – 2019 **Graduate Research Assistant,** Los Alamos National Laboratory
— worked on quantum algorithms for simulation and optimization problems

2018 **Research Intern,** Microsoft Corporation
— developed hybrid quantum-classical algorithms for state preparation with view towards near-term applications.

2017 **Teaching Assistant,** University of New Mexico
— instructor for graduate quantum mechanics; graded homework and assisted students with homework problems.

2016 – 2017 **Student Researcher,** New Mexico Consortium
— investigated quantum algorithms for characterizing and speeding up random walks and Markov processes, with relevance to improving simulation of physical processes.

2013 – 2015 **Teaching Assistant,** University of New Mexico
— instructor for introductory physics lab courses

Awards

- National Science Foundation (NSF) travel award, 2017
- UNM Graduate and Professional Students Association (GPSA) Professional Development Grant, 2015
- Selected for the Visiting Students Program in physics at Harishchandra Research Institute, Allahabad, India, 2012
- Visiting Students Research Program (VSRP), Tata Institute of Fundamental Research, Mumbai, India, 2011
- Recipient of the Innovation in Science Pursuit for Inspired Research (INSPIRE) Scholarship for Higher Education, administered by the Department of Science & Technology, Government of India, 2008–2013

References

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